3. Troubleshooting

Motor Does Not Run

When the motor does not run, refer to P.2-64, "Display of Factor of No-Motor Running" of Preparation as well.

Classification	Causes		Measures
Parameter	Setup of the control mode is not correct	Check that the present control mode is correct with monitor mode of the front panel.	1) Set up Pr0.01 again. 2) Check that the input to control mode switching (C-MODE) of the Cnnector X4 is correct, when Pr0.01 is set to 3 to 5.
	Selection of torque limit is not correct	Check that the external analog input (N-ATL/ P-ATL) is not used for the torque limit.	1) Set up Pr05.21 to 0 and apply -9 [V] to N-ATL and +9 [V] to P-ATL when you use the external input. 2) Set up Pr05.21 to 1 and set up the max. value to Pr0.13 when you use the parameter value.
	Setup of electronic gear is not correct. (Position/Full-closed)	Check that the motor moves by expected revolution against the command pulses.	1) Check the setups of Pr0.09, Pr0.01 and Pr5.00 to Pr5.02 again. 2) Connect the electronic gear switching input (DIV) of Connector X4 to COM–, or invalidate the division/ multiplication switching by setting up the same value to Pr0.09 and Pr5.00.
Wiring	Servo-ON input of Connector X4 (SRV-ON) is open.	In the front panel monitor mode, is the Pin No. corresponding to SRV-ON in " - " state?	Check and make a wiring so as to connect the SRV-ON input to COM–.
	Positive/negative direction over-travel inhibit input of Connector X4 (NOT/POT) is open.	In the front panel monitor mode, is the Pin No. corresponding to NOT/ POT in " A " state?	1) Check and make a wiring so as to connect both NOT/POT inputs to COM–. 2) Set up Pr5.04 to 1 (invalid) and reset the power.
	Command pulse input setup is incorrect. (Position/Full-closed)	Check that the input pulse counts and variation of command pulse sum does not slips, with monitor mode of the front panel.	1) Check that the command pulses are entered correctly to the direction selected with Pr0.05. 2) Check that the command pulses are entered correctly in the format selected with Pr0.07.
	Command pulse input inhibition (INH) of Connector X4 is open. (Position/ Full-closed)	In the front panel monitor mode, is the Pin No. corresponding to INH in " A " state?	1) Check and make a wiring so as to connect the INH input to COM–. 2) Set up Pr5.18 to 1 (invalid).
	Counter clear input (CL) of Connector X4 is connected to COM–. (Position/ Full-closed)	In the front panel monitor mode, is the Pin No. corresponding to CL in " A " state?	1) Check and make wiring so as to open the CL input 2) Set up Pr5.17 to 2 (invalid).

Related page • P.2-52 ... "How to Use the Front Panel" • P.3-30 "Inputs and outputs on connector X4"
• P.4-4... "Details of parameter"

When the motor does not run, refer to P.2-64, "Display of Factor of No-Motor Running" of Preparation as well.

Classification	Causes		Measures
Wiring	Speed command is invalid (Velocity)	Check that the velocity command input method (external analog command/internal velocity command) is correct.	1) Check the setups of Pr3.02 to Pr3.03 again by setting up Pr3.00 to 0, when you use the external analog command. 2) Set up Pr3.04 to Pr3.07 and Pr3.08 to Pr3.11 by setting up Pr3.00 to either one of 1, 2 or 3, when you use the internal speed command.
	Speed zero clamp input (ZEROSPD) of Connecter X4 is open. (Velocity/Torque)	In the front panel monitor mode, is the Pin No. corresponding to ZEROSPD in " A " state?	1) Check and make wiring so as to connect speed zero clamp input to COM–. 2) Set up Pr3.15.
	Torque command is invalid (Torque)	Check that the torque command input method (SPR/TRQR input, P-ATL/TRQR input) is correct.	1) Check that the input voltage is applied correctly by setting up Pr3.17 to 0, when you use SPR/TRQR input. 2) Check that the input voltage is applied correctly by setting up Pr3.17 to 1, when you use the P-ATL/TRQR input.
	Velocity control is invalid (Torque)	Check that the velocity limit input method (parameter velocity, SPR/TRQR/SPL input) is correct.	1) Set up the desired value to Pr3.21 by setting up Pr3.17 to 0, when you use the parameter speed. 2) Check that the input voltage is applied correctly by setting up Pr3.17 to 1, when you use the SPR/TRQR/SPL input.
Installation	Main power is shut off.	In the front panel monitor mode, is the Pin No. corresponding to S-RDY in " - " state?	Check the wiring/voltage of main power of the driver (L1, L2 and L3).
	The motor shaft drags, the motor does not run.	1)Check that you can turn the motor shaft, after turning off the power and separate it from the machine. 2)Check that you can turn the motor shaft while applying DC24V to the brake in case of the motor with electromagnetic brake.	If you cannot turn the motor shaft, consult with the dealer for repair.

Related page • P.4-24... "Details of parameter" • P.3-30 "Inputs and outputs on connector X4"

3. Troubleshooting

Unstable Rotation (Not Smooth), Motor Runs Slowly Even with Speed Zero at Velocity Control Mode

Classification	Causes	Measures
Parameter	Setup of the control mode is not correct.	If you set up Pr0.01 to 1(Velocity control mode) by mistake at position control mode, the motor runs slowly at servo-ON due to speed command offset. Change the setup of Pr0.01 to 0.
Adjustment	Gain adjustment is not proper.	Increase the setup of Pr1.01, 1st velocity loop gain. Enter torque filter of Pr1.04 and increase the setup of Pr1.01 again.
	Velocity and position command are not stable.	Check the motor movement with check pin of the front panel or the waveform graphic function of the PANATERM. Review the wiring, connector contact failure and controller.
Wiring	Each input signal of Connector X4 is chattering. 1) Servo-ON signal 2) Positive/Negative direction torque limit input signal 3) Deviation counter input signal 4) Speed zero clamp signal 5) Command pulse inhibition input	1) Check the wiring and connection between Pin29 and 41 of the Connector X4 using the display function of I/O signal status. Correct the wiring and connection so that the Servo-ON signal can be turned on normally. Review the controller. 2) Check the wiring and connection between Pin-18 and 17, 16 and 17 of the Connector X4 using tester or oscilloscope. Correct the wiring and connection so that Positive/Negative direction torque limit input can be entered normally. 3) Check the wiring and connection between Pin-30 and 41, 16 and 17 of the Connector X4 using display function of I/O signal status. Correct the wiring and connection so that the deviation counter input can be turned on normally. Review the controller. 4) Check the wiring and connection between Pin-26 and 41of the Connector X4 using Display function of I/O signal status. Correct the wiring and connection so that the speed zero clamp input can be entered normally. Review the controller. 5) Check the wiring and connection between Pin-33 and 41of the Connector X4 using display function of I/O signal status. Correct the wiring and connection so that the command pulse inhibition input can be entered normally. Review the controller.
	Noise is on the velocity command.	Use a shield cable for connecting cable to the Connector X4. Separate the power line and signal line (30cm or longer) in the separate duct.
	Slip of offset	Check the voltage between Pin-14 and 15 (speed command input) using a tester or an oscilloscope. Adjust the Pr52 value so that the motor stops.
	Noise is on the position command.	Use a shield cable for connecting cable to the Connector X4. Separate the power line and signal line (30cm or longer) in the separate duct.

Related page • P.4-42... “Details of parameter” • P.3-30 “Inputs and outputs on connector X4”
• P.7-26 “Outline of Setup support software “PANATERM”

3. Troubleshooting

Positioning Accuracy Is Poor

Classification	Causes	Measures
System	Position command is not correct.	Count the feedback pulses with a monitor function of the PANATERM or feedback pulse monitor mode of the console while repeating the movement of the same distance. If the value does not return to the same value, review the controller. Make a noise measure to command pulse.
	Captures the positioning complete signal at the edge.	Monitor the deviation at positioning complete signal reception with the Connector X7 or the waveform graphic function of the PANATERM. Make the controller capture the signal not at the edge but with some time allowance.
	Shape or width of the command pulse is not per the specifications.	If the shape of the command pulse is broken or narrowed, review the pulse generating circuit. Make a noise measure.
	Noise is superposed on deviation counter clear input CL (Connector X4, Pin-5).	Make a noise measure to external DC power supply and make no wiring of the unused signal lines.
Adjustment	Position loop gain is small.	Check the position deviation with the monitor function of the PANATERM or at the monitor mode of the console. Increase the setup of Pr1.00 within the range where no oscillation occurs.
Parameter	Setup of the positioning complete range is large.	Lower the setup of Pr4.31 within the range where no chattering of complete signal occurs.
	Command pulse frequency have exceeded 500kpps or 4Mpps.	Lower the command pulse frequency. Change the division/ multiplication ratio of 1st and 2nd numerator of command division/ multiplication, Pr0.09 and Pr0.10. Use a pulse line interface exclusive to line driver when pulse line interface is used.
	Setup of the division/ multiplication is not correct.	Check if the repetition accuracy is same or not. If it does not change, use a larger capacity motor and driver.
	Velocity loop gain is proportion action at motor in stall.	• Set up Pr1.02 and Pr1.07 of time constant of velocity loop integration to 9999 or smaller. • Review the wiring and connection so that the connection between Pin-27 and 41 of the gain switching input connector, Connector X4 becomes off while you set up Pr1.14 of 2nd gain setup, to 1.
Wiring	Each input signal of Connector X4 is chattering. 1) Servo-ON signal 2) Deviation counter clear input signal 3) Positive/Negative direction torque limit input signal 4) Command pulse inhibition input	1) Check the wiring and connection between Pin29 and 41 of the connector, Connector X4 using the display function of I/O signal status. Correct the wiring and connection so that the servo-On signal can be turned on normally. Review the controller. 2) Check the wiring and connection between Pin-30 and 41, 16 and 17 of the connector, Connector X4 using display function of I/O signal status. Correct the wiring and connection so that the deviation counter clear input can be turned on normally. Review the controller. 3 Check the wiring and connection between Pin-18 and 17, 16 and 17 of the connector, Connector X4 using tester or oscilloscope. Correct the wiring and connection so that Positive/Negative direction torque limit input can be entered normally. 4) Check the wiring and connection between Pin-33 and 41of the connector, Connector X4 using display function of I/O signal status. Correct the wiring and connection so that the command pulse inhibition input can be entered normally. Review the controller.
Installation	Load inertia is large.	Check the overshoot at stopping with graphic function of the PANATERM. If no improvement is obtained, increase the driver and motor capacity.

Related page • P.4-4 “Details of parameter” • P.3-30 “Inputs and outputs on connector X4”
• P.7-26 “Outline of Setup support software “PANATERM”

3. Troubleshooting

Origin Point Slips

Classification	Causes	Measures
System	Z-phase is not detected.	Check that the Z-phase matches to the center of proximity dog. Execute the homing matching to the controller correctly.
	Homing creep speed is fast.	Lower the homing speed at origin proximity. Or widen the origin sensor.
Wiring	Chattering of proximity sensor (proximity dog sensor) output .	Check the dog sensor input signal of the controller with oscilloscope. Review the wiring near to proximity dog and make a noise measure or reduce noise.
	Noise is on the encoder line.	Reduce noise (installation of noise filter or ferrite core), shield treatment of I/F cables, use of a twisted pair or separation of power and signal lines.
	No Z-phase signal output.	Check the Z-phase signal with oscilloscope. Check that the Pin-13 of the connector, CN X5 is connected to the earth of the controller. Connect the earth of the controller because the open collector interface is not insulated. Replace the motor and driver. Request for repair.
	Miswiring of Z-phase output.	Check the wiring to see only one side of the line driver is connected or not. Use a CZ output (open collector if the controller is not differential input.

6
When in Trouble

3. Troubleshooting

Abnormal Motor Noise or Vibration

Classification	Causes	Measures
Wiring	Noise is on the speed command.	Measure the speed command inputs of Pin-14 and 15 of the connector, Connector X4 with an oscilloscope. Reduce noise (installation of noise filter or ferrite core), shield treatment of I/F cables, use of a twisted pair, separation of power and signal lines.
Adjustment	Gain setup is large.	Lower the gain by setting up lower values to Pr1.01 and 1.09, of velocity loop gain and Pr1.00 and Pr1.05 of position loop gain.
Installation	Resonance of the machine and the motor.	Re-adjust Pr1.04 and 1.09 (Torque filter). Check if the machine resonance exists or not with frequency characteristics analyzing function of the PANATERM. Set up the notch frequency to Pr2.01, Pr2.04, Pr2.07 or Pr2.10 if resonance exists.
	Motor bearing	Check the noise and vibration near the bearing of the motor while running the motor with no load. Replace the motor to check. Request for repair.
	Electro-magnetic sound, gear noise, rubbing noise at brake engagement, hub noise or rubbing noise of encoder.	Check the noise of the motor while running the motor with no load. Replace the motor to check. Request for repair.

3. Troubleshooting

Overshoot/Undershoot, Overheating of the Motor (Motor Burn-Out)

Classification	Causes	Measures
Adjustment	Gain adjustment is not proper.	Check with graphic function of PANATERM or velocity monitor (SP) or torque monitor (IM). Make a correct gain adjustment. Refer to “Adjustment”.
Installation	Load inertia is large.	Check with graphic function of PANATERM or monitor (Connector X7). Make an appropriate adjustment. Increase the motor and driver capacity and lower the inertia ratio. Use a gear reducer.
	Looseness or slip of the machine.	Review the mounting to the machine.
	Ambient temperature, environment.	Lower the temperature with cooling fan if the ambient temperature exceeds the predications.
	Stall of cooling fan, dirt of fan ventilation duct.	Check the cooling fans of the driver and the machine. Replace the driver fan or request for repair.
	Mismatching of the driver and the motor.	Check the name plates of the driver and the motor. Select a correct combination of them referring to the instruction manual or catalogue.
	Failure of motor bearing.	Check that the motor does not generate rumbling noise while turning it by hand after shutting off the power. Replace the motor and request for repair if the noise is heard.
	Electromagnetic brake is kept engaged (left un-released).	Check the voltage at brake terminals. Apply the power (DC24V) to release the brake.
	Motor failure (oil, water or others)	Avoid the installation place where the motor is subject to high temperature, humidity, oil, dust or iron particles.
	Motor has been turned by external force while dynamic brake has been engaged.	Check the running pattern, working condition and operating status, and inhibit the operation under the condition of the left.

6 When in Trouble

3. Troubleshooting

Motor Speed Does Not Reach to the Setup, Motor Revolutions (Travel) Is Too Large or Small

Classification	Causes	Measures
Parameter	Velocity command input gain is not correct.	Check that the setup of Pr3.02, speed command input gain, is made so as to make the setup of 500 makes 3000 r/min.
Adjustment	Position loop gain is low.	Set up Pr1.00and Pr1.05, position loop gain to approx. 1000.
	Division/Multiplication is not proper.	Set up correct values to Pr0.09, 1st numerator of electronic gear, Pr0.11, numerator multiplier of electronic gear and Pr0.10, denominator of electronic gear. Refer to parameter setup at each mode.

Related page • P.4-4 “Details of parameter” • P.7-26 “Outline of Setup support software “PANATERM”

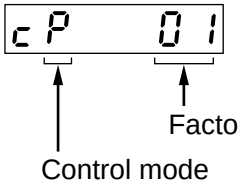
3. Troubleshooting

Parameter Returns to Previous Setup

Classification	Causes	Measures
Parameter	No writing to EEPROM has been carried out before turning off the power.	Refer to P.2-71, “How to Operate-EEPROM Writing” of Preparation.

10) Display of the Factor of No-Motor Running

Displays the factor of no-motor running in number.



P .....Position control
t .....Torque control
S ..... Velocity control
F ..... Full-closed control

• Explanation of factor No.

Factor No.	Factor	Related Control Mode				Content
		P	S	T	F	
flashing	Occurrence of error/alarm	P	S	T	F	An error is occurring, and an alarm is triggered.
00	No particular factor	P	S	T	F	No factor is detected for No-motor run. The motor runs in normal case.
01	Main power shutoff	P	S	T	F	The main power of the driver is not turned on.
02	No entry of SRV-ON input	P	S	T	F	The Servo-ON input (SRV-ON) is not connected to COM–.
03	Over-travel inhibition input is valid	P	S	T	F	While Pr5.04 is 0 (Run-inhibition input is valid), • Positive direction over-travel inhibition input (POT) is open and speed command is Positive direction. • Negative direction over-travel inhibition input (NOT) is open and speed command is Negative direction.
04	Torque limit setup is small	P	S	T	F	Either one of the valid torque limit setup value of Pr0.13 (1st) or Pr5.22 (2nd) is set to 5% or lower than the rating.
05	Analog torque limit input is valid.	P	S	T	F	While Pr5.21 is 0 (analog torque limit input accepted), • Positive direction analog torque limit input (P-ATL) is negative voltage and speed command is Positive direction. • Negative direction analog torque limit input (N-ATL) is positive voltage and speed command is Negative direction.
06	INH input is valid.	P	S	T	F	Pr5.18 is 0 (Command pulse inhibition input is valid.), and INH is open.
07	Command pulse input frequency is low.	P	S	T	F	The position command per each control cycle is 1 pulse or smaller due to, • No correct entry of command pulse • No correct connection to the input selected with Pr0.05. • No matching to input status selected with Pr0.06 pr Pr0.07.
08	CL input is valid.	P	S	T	F	While Pr5.17 is 0 (Deviation counter clear at level), the deviation counter clear input (CL) is connected to COM–.
09	ZEROSPD input is valid.	P	S	T	F	While Pr3.15 is 1 (Speed zero clamp is valid.), the speed zero clamp input (ZEROSPD) is open.
10	External speed command is small.	P	S	T	F	While the analog speed command is selected, the analog speed command is smaller than 0.06[V].
11	Internal speed command is 0.	P	S	T	F	While the internal speed command is selected, the internal speed command is set to lower than 30 [r/min]
12	Torque command is small.	P	S	T	F	The analog torque command input (SPR or P-ATL) is smaller than 5 [%] of the rating.
13	Speed limit is small.	P	S	T	F	• While Pr3.17 is 0 (speed is limited by 4th speed of internal speed), Pr3.07, (4th speed of speed setup) is set to lower than 30 [r/min]. • While Pr3.17 is 1 (speed is limited by SPR input), the analog speed limit input (SPR) is smaller than 0.06 [V].
14	Other factor	P	S	T	F	The motor runs at 20 [r/min] or lower even though the factors from 1 to 13 are cleared, (the command is small, the load is heavy, the motor lock or hitting, driver/motor fault etc.)

Note * Motor might run even though the other number than 0 is displayed.
Refer to "6.In trouble".